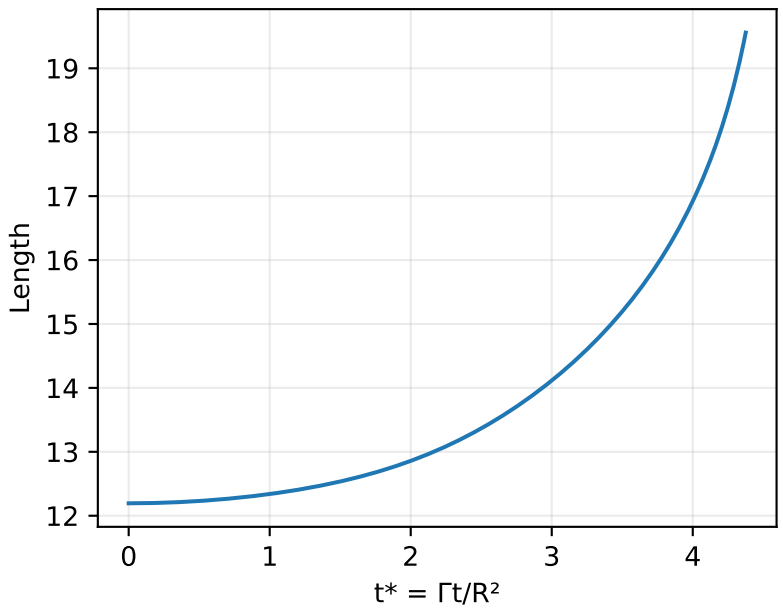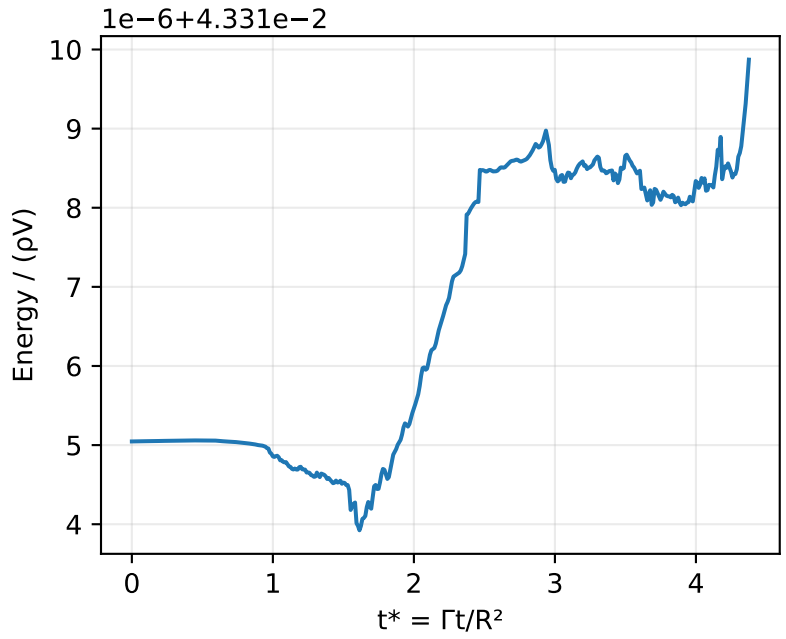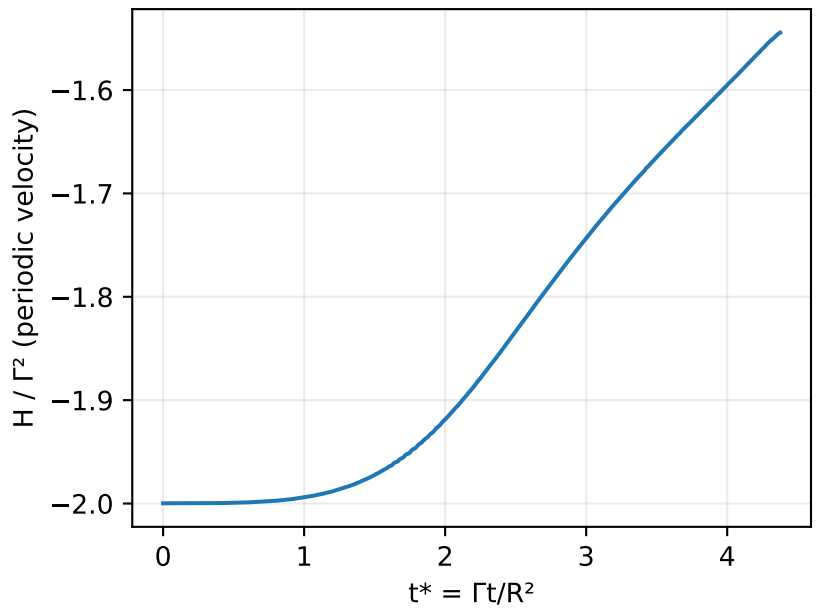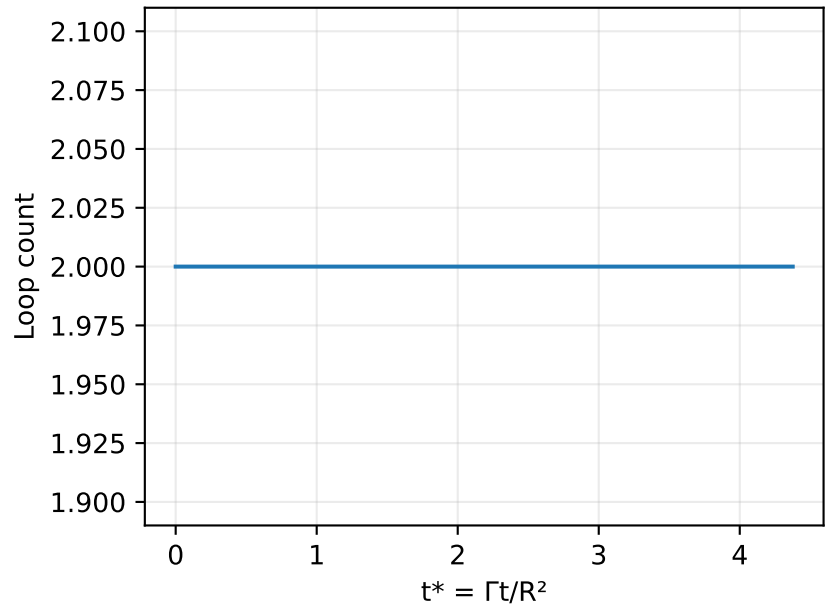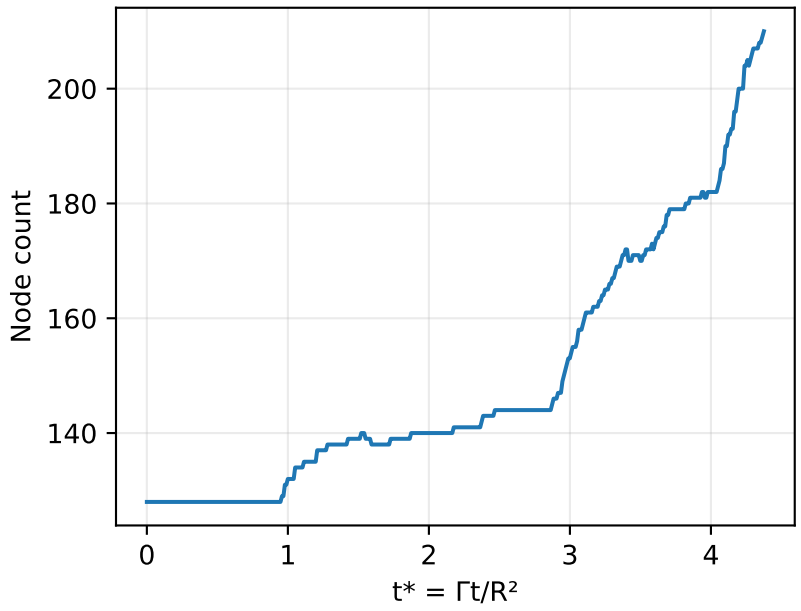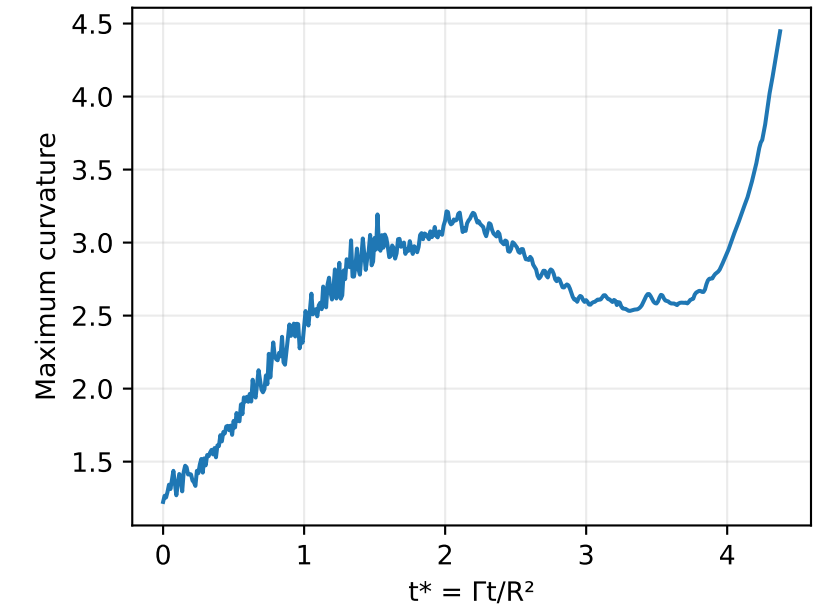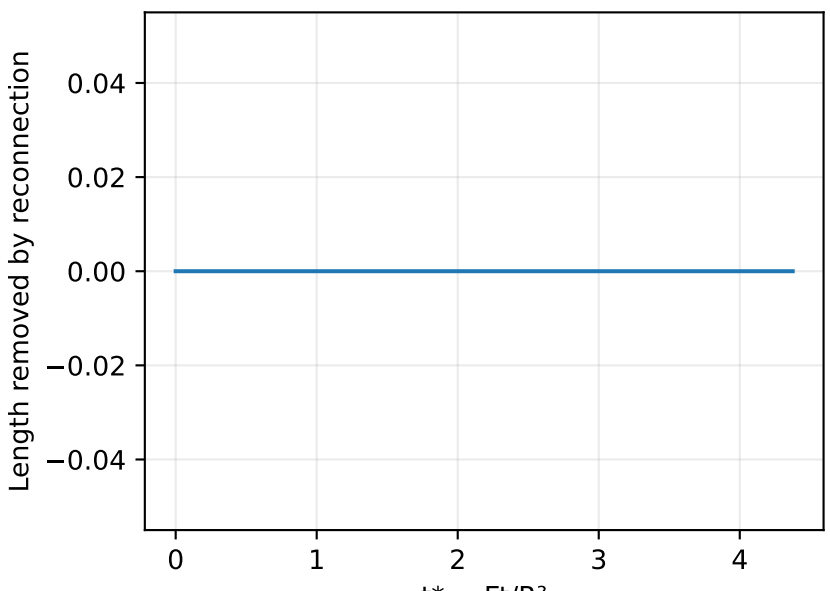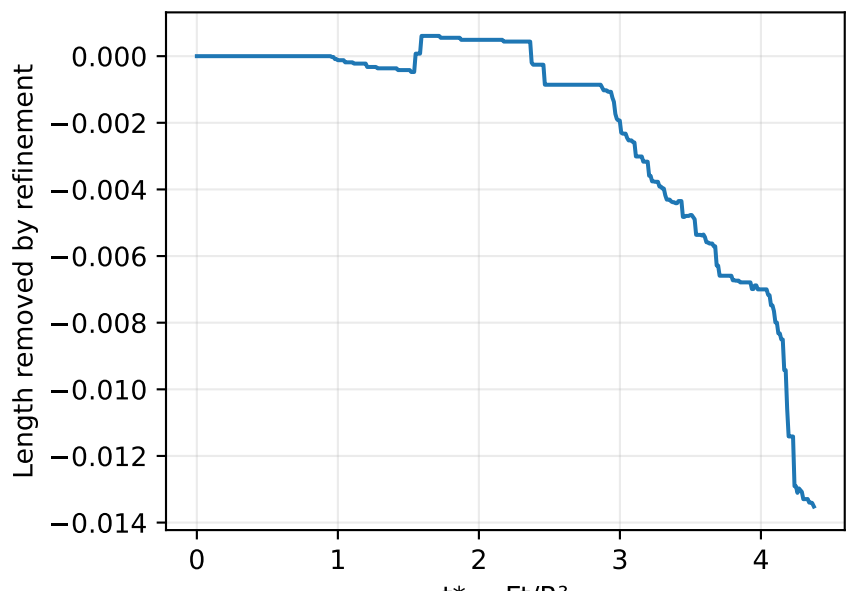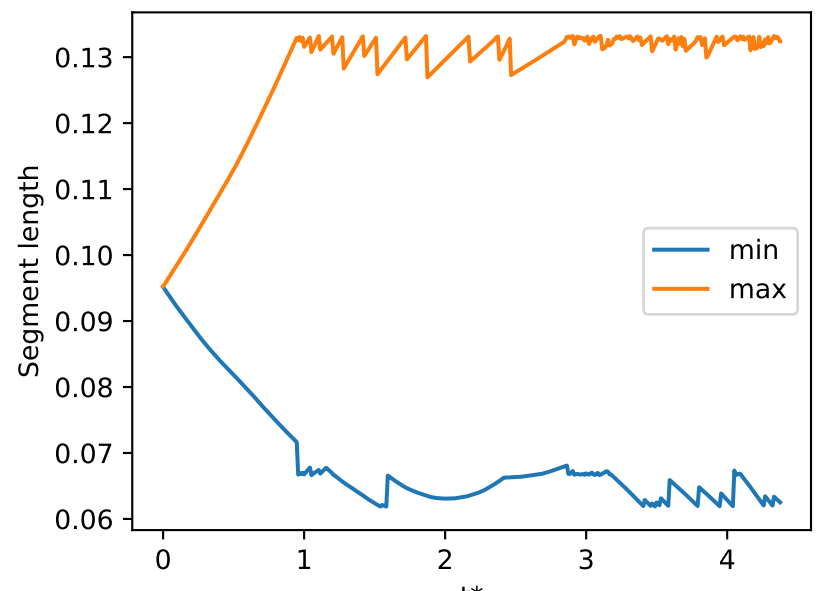hopf_prepared_ref · a=0.1171875 · no mutual friction